# Submanifolds

## Normal Forms, Immersions, Submersions & Regular Submanifolds

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# Normal Form of an Immersion

# Immersions have a local standard form

**Theorem (Normal form of an immersion)**

*Let $f : M^m \longrightarrow N^n$ be an immersion. For every $p \in M$, there exists a local coordinate system $(U, \varphi)$ about $p$ and a local coordinate system $(V, \psi)$ about $q = f(p)$, s.t. the local form of $f$*

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)$$

*is a standard embedding, i.e.*

$$\psi \circ f \circ \varphi^{-1}(x^1, x^2, \ldots, x^m) = (x^1, x^2, \ldots, x^m, 0, \ldots, 0).$$

**Idea**

An immersion looks, near each point, like the inclusion of a coordinate subspace $\mathbb{R}^m \hookrightarrow \mathbb{R}^n$.

# Immersions are locally embeddings

**Corollary**

*An immersion map $f : M \longrightarrow N$ is an embedding locally: around every point $p \in M$, there is a neighborhood $U \subset M$ such that $f\big|_U : U \to f(U)$ is a homeomorphism onto its image.*

**Remark**

Globally an immersion need *not* be an embedding — the image may cross itself (e.g. the figure-eight curve). The local normal form only controls a small neighborhood of each point.

# Normal Form of a Submersion

# Submersions have a local standard form

**Theorem (Normal form of a submersion)**

*Let $f : M^m \longrightarrow N^n$ be a submersion ($m \geq n$). Then $\forall p \in M$, there is a local coordinate system $(U, \varphi)$ of $p$ and $(V, \psi)$ of $q = f(p)$, s.t. the local form*

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)$$

*is a standard submersion, i.e.*

$$\psi \circ f \circ \varphi^{-1}(x_1, \ldots, x_m) = (x_1, \ldots, x_n).$$

**Idea**

A submersion looks, near each point, like the projection $\mathbb{R}^m \twoheadrightarrow \mathbb{R}^n$ onto the first $n$ coordinates.

# Submersions are open maps

**Corollary**

*A submersion $f : M \longrightarrow N$ is an open map, i.e. it maps open sets to open sets.*

**Proof.**

Let $W \subset M$ be open and $p \in W$. In the normal-form coordinates above, $f$ is a coordinate projection, which is open on $\varphi(U \cap W)$; hence $f(U \cap W)$ is a neighborhood of $f(p)$ contained in $f(W)$. Since $p \in W$ is arbitrary, $f(W)$ is open. □

**Remark**

In particular, a *surjective* submersion $f : M \to N$ is a quotient map: $V \subset N$ is open iff $f^{-1}(V) \subset M$ is open.

# Submanifolds

# Two notions of submanifolds

**Definition (Submanifolds)**

Let $M, N$ be $C^\infty$-manifolds with $M \subset N$ as a set. If the inclusion map $i : M \hookrightarrow N$ is an immersion, then $M$ is an *immersion submanifold* (or submfd). If $i : M \hookrightarrow N$ is an embedding (i.e. $M \subset N$ as a topological subspace), then $M$ is called a *regular submanifold*.

**Lemma**

*Let $M$ and $N$ be $C^\infty$-manifolds. If $f : M \longrightarrow N$ is a one-to-one immersion, then $f(M) \subset N$ is an immersion submanifold (with the topological structure inherited from $M$). If $f : M \longrightarrow N$ is an embedding, then $f(M) \subset N$ is a regular submanifold.*

# Immersion submanifold vs. regular submanifold

## Terminology (following the lemma)

When $f : M \longrightarrow N$ is a one-to-one immersion, we call $M$ (or its image) an *immersion submanifold*. If $f$ is an embedding, we call $M$ a *regular submanifold*.

## Key distinction

- **Immersion submanifold:** the topology on $M$ may be *finer* than the subspace topology inherited from $N$; "short" pieces of $M$ may be far apart in $N$.
- **Regular submanifold:** $M$ carries exactly the subspace topology; locally $M$ looks like a *slice* of $N$ — see the structure theorem below.

# The Structure of Regular Submanifolds

# Regular submanifolds are locally slices

**Theorem (The structure of regular submanifolds)**

*$M$ is a regular submanifold of $N \iff$ $M$ is a sub-topological space of $N$, and $\forall p \in M$, there is a local coordinate system $(U, x_1, \ldots, x_n)$ of $p$ in $N$, s.t.*

$$M \cap U = \left\{ q \in U \;\middle|\; x_i(q) = 0,\ m+1 \leq i \leq n \right\}.$$

**Remark**

$\{x_i\}_{m+1 \leq i \leq n}$ are called *slice coordinates*: locally $M$ is cut out by setting the last $n - m$ coordinates to zero, and $(x_1, \ldots, x_m)\big|_M$ give a chart of $M$.

# The constant rank theorem

## Theorem (Constant rank map)

*Let $f : M^m \longrightarrow N^n$ be a $C^\infty$ map with*

$$\operatorname{rank}_p f \equiv \text{const} = l \qquad (\forall p \in M).$$

*Then $\forall q \in N$, the level set*

$$f^{-1}(q) = \{p \in M \mid f(p) = q\}$$

*is either $\varnothing$, or a regular submanifold of $M$ of dimension $(m - l)$.*

## Special cases

- $l = n$ (submersion): $f^{-1}(q)$ is a regular submanifold of dimension $m - n$ — the preimage theorem.
- $l = m$ (immersion, $m \leq n$): $f^{-1}(q)$ is a 0-dimensional submanifold (discrete points).

# Proof of the Structure Theorem

# Setup of the proof ($\Rightarrow$)

Let $M$ be a regular submanifold of $N$. The inclusion map

$$i : M \longrightarrow N$$

is an embedding, where $M$ is endowed with the subspace topology. The differential

$$i_{*,p} : T_pM \longrightarrow T_pN$$

has $\mathbf{rank}\, i_{*,p} = m$ ($m = \mathbf{dim}\, M$). Let $p \in M$; take two local coordinate systems about $p$:

$$(U, \varphi = (x_1, \ldots, x_m)) \text{ on } M, \qquad (V, \psi = (y_1, \ldots, y_n)) \text{ on } N.$$

# Full rank of the Jacobian

Then the Jacobian

$$\left.\frac{\partial\left(y_i \circ \varphi^{-1}\right)}{\partial x_j}\right|_{\substack{1\leq i\leq n\\ 1\leq j\leq m}} \left(\varphi(p)\right) \quad \text{has full rank } m.$$

WLOG (relabeling the $y_i$'s), we assume the $m \times m$ submatrix

$$\left.\frac{\partial\left(y_i \circ \varphi^{-1}\right)}{\partial x_j}\right|_{\substack{1\leq i\leq m\\ 1\leq j\leq m}} \left(\varphi(p)\right) \quad \text{is non-singular.}$$

# Extending to a local diffeomorphism

Consider the new map $\Phi : \varphi(U) \times \mathbb{R}^{n-m} \longrightarrow \psi(V)$,

$$(x_1, \ldots, x_m; x_{m+1}, \ldots, x_n)$$
$$\longmapsto \big(y_1 \circ \varphi^{-1}, \ldots, y_m \circ \varphi^{-1}, y_{m+1} \circ \varphi^{-1} + x_{m+1}, \ldots, y_n \circ \varphi^{-1} + x_n\big).$$

Then

$$D\Phi\Big|_{(\varphi(p),\,0)} = \begin{pmatrix} \left(\dfrac{\partial y_i \circ \varphi^{-1}}{\partial x_j}\right)_{m\times m} & 0 \\ * & I_{n-m} \end{pmatrix}$$

is non-singular, so $\Phi$ is a *local diffeomorphism* about $(\varphi(p), 0)$.

# Conclusion of the proof

Consider

$$\Phi^{-1} \circ \psi \circ i \circ \varphi^{-1}(x_1, \ldots, x_m) = (x_1, \ldots, x_m,\ 0, \ldots, 0).$$

Indeed, by the definition of $\Phi$, on the slice $(x_{m+1}, \ldots, x_n) = (0, \ldots, 0)$ we have $\Phi \circ (\psi \circ i \circ \varphi^{-1})(x_1, \ldots, x_m) = (x_1, \ldots, x_m, 0, \ldots, 0)$. Hence the new coordinates on $N$

$$\widetilde{x}_i \triangleq (\Phi^{-1} \circ \psi)_i, \qquad i = 1, \ldots, n,$$

satisfy

$$M \cap U = \left\{ q \in U \;\middle|\; \widetilde{x}_i(q) = 0,\ m+1 \le i \le n \right\}. \quad \square$$